

MACHINE LEARNING PROJECT REPORT

NewsLens: News Category Classification Using a Neural Network

An end-to-end text classification system with automated preprocessing, a neural network, FastAPI, React, and Render deployment.

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Abstract

NewsLens automatically classifies news headlines and article excerpts into World, Sports, Business, and Sci/Tech. The system combines automated text preprocessing, TF-IDF feature extraction, a feed-forward neural network, a FastAPI prediction service, and a professional React interface. A Streamlit interface and Render deployment configuration are also included.

Project snapshot: 32 balanced local training articles, 958 TF-IDF features, 4 output categories, and a two-layer MLP classifier.

1. Introduction

The growth of digital news makes manual organization difficult. This project develops an end-to-end classifier that accepts a headline or short article excerpt and predicts the most likely category.

World

International affairs, diplomacy, disasters, and global events.

Sports

Competitions, athletes, teams, and tournaments.

Business

Markets, companies, finance, trade, and economics.

Sci/Tech

Science, software, devices, research, and technology.

2. Objectives

1. Build automated news text preprocessing.
2. Convert cleaned text into TF-IDF features.
3. Train a multiclass neural network.
4. Display predictions and ranked confidence signals.
5. Provide a usable web interface.
6. Package the system for local and Render deployment.

3. System Architecture

```
User headline
|
React frontend -- POST /api/predict --> FastAPI
|
```

```
clean text + TF-IDF
      |
    MLP classifier
      |
category + confidence signals
```

The data layer contains an AG News-style CSV. The ML layer performs cleaning, vectorization, and classification. FastAPI exposes health and prediction endpoints. The React interface displays the result.

4. Dataset

The local training file is `data/train.csv` and contains a balanced demonstration dataset.

Category	Articles
World	8
Sports	8
Business	8
Sci/Tech	8
Total	32

A larger AG News dataset can be substituted for stronger generalization.

5. Automated Preprocessing

Every dataset load combines title and description, decodes HTML, removes tags and punctuation, converts text to lowercase, removes accents, collapses whitespace, removes missing or duplicate records, validates the four labels, and writes `data/preprocessed_news.csv`.

6. Feature Engineering

TF-IDF represents each article numerically. The implementation uses unigram and bigram features with sublinear term frequency. The current local training run produces **958 TF-IDF features**.

7. Neural Network Model

- Input: TF-IDF feature vector
- Hidden layer 1: 64 neurons
- Hidden layer 2: 32 neurons
- Output: four category probabilities

- Maximum iterations: 250
- Random state: 42

The model returns probabilities for all categories. The application displays the highest-probability category and the remaining categories as alternative signals.

8. Application Features

React interface

The React frontend uses a neon control-room design with model status, headline input, example buttons, confidence bars, alternative signals, and responsive layouts.

FastAPI service

`GET /api/health` reports model status and statistics. `POST /api/predict` accepts text and returns the predicted category with confidence values. In production, FastAPI serves the built React files from the same origin.

Streamlit interface

The Streamlit version supports CSV upload, automatic preprocessing, model inspection, and rapid experimentation.

9. Example Prediction

Input: Scientists develop a battery that charges electric cars in five minutes

Expected category: Sci/Tech

Confidence is a model output and should not be interpreted as a guarantee of factual correctness.

10. Deployment

The included `render.yaml` installs Python and React dependencies, builds the Vite frontend, starts Uvicorn on Render's assigned port, serves the React build through FastAPI, and uses `/api/health` as the health check. The `start_news1ens.bat` launcher starts both local services and opens the browser.

11. Limitations

- The local dataset is small and intended for demonstration.
- Small datasets can produce overconfident predictions.
- The system does not verify facts or detect misinformation.
- Formal accuracy, precision, recall, F1-score, and confusion-matrix evaluation should be added with a larger dataset.

12. Future Enhancements

1. Train on the complete AG News dataset.
2. Add train, validation, and test evaluation.
3. Compare Logistic Regression, SVM, and transformer models.
4. Add confidence calibration and low-confidence warnings.
5. Store prediction history and export results.
6. Add authentication, rate limiting, and automated tests.

13. Conclusion

NewsLens demonstrates a complete workflow from raw CSV data to a deployed news classification application. Automated preprocessing improves repeatability, TF-IDF provides a practical text representation, and the neural network produces four category predictions. The React and FastAPI architecture makes the model accessible through a professional interface, while Render configuration supports straightforward hosting.

References

1. AG News dataset format and four-category classification task.
2. Scikit-learn documentation for TfidfVectorizer and MLPClassifier.
3. FastAPI documentation.
4. React and Vite documentation.
5. Render web-service documentation.